

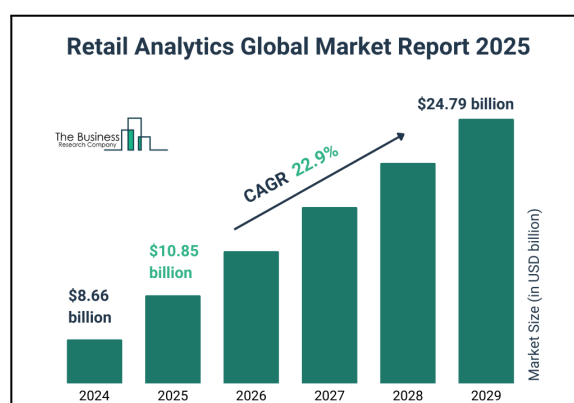
Data Analytics in the Cloud MBD-EN2025 APRIL

Group Work - Option B: Cloud Data Platform for Retail

Scenario

Retail is among the most rapidly changing industries in recent years, driven by globalization and the exponential growth of data. As consumers engage across physical and digital touchpoints, retailers must manage massive volumes of data in real time, requiring scalable and flexible infrastructure. As a result, the retail analytics market size has increased exponentially over the past few years, and it is expected to grow to \$24.79 billion in 2029, at a compound annual growth rate (CAGR) of 22.9%¹.

Companies like Walmart² and IKEA³ are leveraging cloud platforms to centralize their data, enabling advanced analytics for demand forecasting, inventory optimization, and personalized customer experiences.



Option B of the Group Work for this edition of the Data Analytics in the Cloud course will focus on a retail case. Specifically, the proposed scenario is loosely (but quite obviously) inspired by the Spanish fashion retailer Mango. A few years ago, Mango migrated part of its workloads to AWS cloud in collaboration with the certified partner NTT Data, as you can read in the case study⁴.

In order to evaluate different professional and technological solutions in a comparable and objective manner, issuing a Request for Proposal (RFP) is considered a sound business practice. Providers aiming to win the project—and with it, the associated financial and

¹ [Precedence Research - Retail Analytics Market Size, Share and Trends 2024 to 2034](#)

² [Walmart - Walmart establishes strategic partnership with Microsoft to further accelerate digital innovation in retail](#)

³ [Retail Touchpoints - IKEA Partners With Google Cloud to Enhance Omnichannel and Customer Service Capabilities](#)

⁴ [NTT Data - Mango Case Study](#)

reputational benefits—must respond to the RFP in detail, competing with other vendors by outlining their proposed solution, pricing, timeline, and relevant experience.

To gain hands-on experience with this type of document in the context of cloud and data analytics, each group will take on the role of a company competing in a project to migrate and modernize the data architecture of a retail business.

Materials

Alongside this group project presentation, you will find three appendices containing the necessary materials:

- **Appendix I – Request for Proposal:** Although in a simplified and more compact format, it contains the typical sections of an RFP. Read all the information carefully, including the instructions for submitting responses.
- **Appendix II – Customer Research:** Includes information that goes beyond what is typically found in an RFP, but is equally valuable for shaping your proposal. You can assume this material was gathered by your group's Business Development team through data mining and client interviews.
- **Appendix III – Sample Datasets:** Synthetic data samples from various existing customer systems, that can be used as a reference to identify data types and potential use cases. You are not expected to analyze or process this data. Again, you can assume this information was shared by the customer, either as an attachment to the RFP or during a conversation with their technical team.

Deliverables

In the Group Work activity of the course platform, each team (3-5 members) must deliver **by Feb 27, 2026 (EoD):**

1. A document with the response to the RFP, following the suggested structure.
2. An Appendix to the RFP document, including:
 - References
 - Generative AI acknowledgement, in accordance with the course guidelines.
 - Contribution of each member to the work.
3. If the team cannot present their project in class, they must provide a ~10-15 min video with the presentation.

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It is not mandatory to use AWS as the cloud service destination; other solutions can be proposed. However, students may use the Capstone Project module in the AWS Academy – IE – Data Analytics in the Cloud as a playground to explore possible architectures. Be cautious with the time and services used to avoid exhausting the system's credit.

Note that delivering a functioning solution is not a requirement of the Group Work.

Evaluation

The group project will be graded on a scale from 0 to 100, contributing proportionally to 30% of the final course grade.

Deliverable		Points
Project document	Completeness & Accuracy All points are addressed with sound responses.	20
	Clarity & Impact Responses are effective and engaging.	20
	Innovation Ideas are original and creative.	10
Video/Live presentation*		30
Individual contribution		20
TOTAL		100

***IMPORTANT:** All team members MUST participate in the video/live presentation.

Appendix 1. Request for Proposal

Migration and modernization of Manga Data Architecture to the Cloud

About Manga



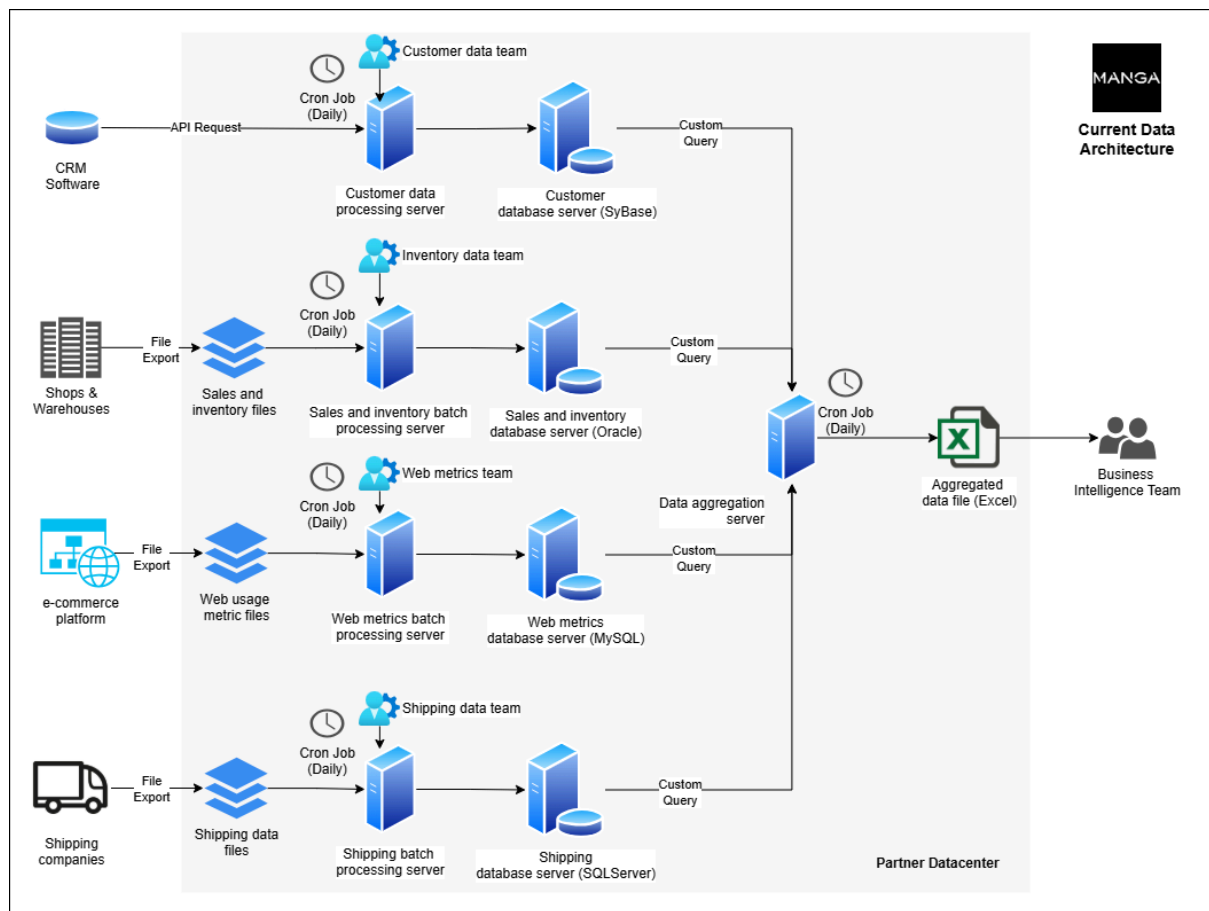
Manga is a Spanish fashion retailer founded in Barcelona in 1999. Since its inception, our brand has experienced steady growth and today operates over 100 physical stores worldwide. Alongside this brick-and-mortar success, Manga has developed a growing e-commerce presence, positioning itself as a competitive player in the global retail space.

As part of our next phase of growth, Manga is planning an ambitious international expansion targeting new customer segments, while also reinforcing its footprint across digital platforms. The company recognizes the strategic importance of leveraging data to deliver personalized experiences, improve operational efficiency, and support business innovation.

Project and Goals

Manga's current data infrastructure is deployed in dedicated servers of our partner data centers (colocation model). The data architecture is built around a series of independent ETL pipelines triggered daily with cron jobs. These pipelines extract, process, and transform data from various sources, including customer information from CRM software, inventory and sales data from physical shops and warehouses, online activity such as visits, transactions, and customer reviews from the web, as well as distribution and return data from external shipping partners. The ETL processes for each pipeline are developed by a different team with their own technical stack, while deployment and maintenance are handled by our partner. Once per day, another cron job aggregates the outputs from these pipelines into a collection of Excel files, which are then shared with BI users for manual analysis and reporting purposes.

The following diagram illustrates Manga's current (as-is) architecture:



Source: Generated with Draw.io

To support our evolving business needs, Manga is actively developing both customer-facing and internal features such as:

- Fast and accurate product recommendations
- Dynamic pricing mechanisms
- Real-time sales forecasting
- Supply chain optimization with smart labels
- Virtual shopping assistant
- Data exposure to external consumers, including AI agents

These initiatives require a modern, scalable, high-performance, and cost-efficient data platform capable of handling both structured and unstructured data, processing in batch and real-time, and serving BI, analytics, and machine learning teams with high-quality data and timely insights.

Scope and Requirements

This initiative is not intended as a simple lift-and-shift migration of the current infrastructure. Instead, Manga seeks a comprehensive modernization of its data platform—an architecture that is cloud-native, agile, future-proof, and extensible to support upcoming business scenarios and innovations. The main project requirements are described below:

R1. Unified multi-modal data platform: Provide a centralized data platform enabling integrated access across departments and business units. Support high-throughput, low-latency streaming data pipelines alongside traditional batch processing. Capable of storing, processing, and querying structured, semi-structured, and unstructured data (e.g., logs, images, clickstream, and social data).

R2. Multi-environment setup: Three separate environments must be provisioned and configured: Development (Dev) for testing and experimentation, Pre-Production (Pre-Prod) for staging and validation, and Production (Prod) for live operations. Anonymized samples of production data should be utilized for the development and pre-production environments to maintain realism while ensuring compliance with privacy standards.

R3. Automation: Key processes must be automated, including infrastructure creation and scaling, data lifecycle management (ingestion, processing, storage, and archiving), and end-to-end data processing workflows. Infrastructure-as-Code (IaC) tools should be leveraged for consistency and repeatability in these processes.

R4. Security and high availability: Advanced security measures must be implemented to protect sensitive data, including encryption, role-based access control, and threat detection. High availability must be ensured with a target uptime of at least 99.9%, supported by redundant systems and failover mechanisms.

R5. Data quality: Data quality monitoring and validation protocols should be established to ensure accuracy, consistency, and completeness. Automated checks and real-time alerts must be implemented to promptly identify and resolve data anomalies.

R6. Cost optimization: The project should achieve measurable cost reductions compared to the existing on-premises platform. Resource allocation must be optimized to minimize operational expenses without compromising performance or security.

R7. Extensibility and interoperability: The platform must enable seamless integration of new components and services without disrupting existing workflows. It should allow easy connection with external consumers such as APIs, third-party platforms, and partner

systems. Backwards compatibility must be ensured across data schemas, APIs, and models to maintain stability during updates.

R8. Sustainability: Environmentally sustainable practices should be adopted in cloud operations by leveraging energy-efficient services and regions. Metrics should be provided to demonstrate improvements in sustainability compared to the on-premises infrastructure.

Structure for Proposal Responses

Proposals must adhere to the following structure to ensure clarity, consistency, and ease of evaluation. The entire response should not exceed 15 pages:

1. **Executive Summary:** Provide a brief overview of your proposed solution, highlighting how it addresses Manga's goals of modernizing its data platform, enabling innovation, and supporting business growth. Summarize the value proposition, key differentiators, and expected benefits for both short- and long-term outcomes.
2. **High-Level Architecture:** Present a conceptual diagram of the proposed architecture, showing the main components of the data engineering lifecycle and their data flow relationships. Do not specify particular technologies at this stage. Instead, explain the architectural decisions based on foundational cloud analytics principles such as scalability, elasticity, modularity, and separation of concerns.
3. **Low-Level Architecture:** Provide a detailed breakdown of the high-level architecture using specific services and tools from your proposed tech stack.. Include a rationale for each component selection based on vendor best practices, performance benchmarks, operational reliability, and cost-effectiveness. Show how each part contributes to the goals of extensibility, interoperability, and real-time data access.
4. **Project Implementation:** Explain how the proposed solution meets the scope and requirements outlined in the RFP. It should articulate the approach, processes, and methodologies employed to achieve the stated objectives and provide a detailed explanation of the value delivered by the solution.
5. **Sample Use Cases:** Demonstrate how the proposed architecture and services address the Manga's data sources and use cases. These examples should cover data ingestion, processing workflows, visualization dashboards, and predictive analytics outputs, showcasing their effectiveness and alignment with the project goals.

6. **Future Improvements:** Propose potential enhancements to the initial proposal, identifying areas for growth and innovation. It should detail the anticipated benefits of these improvements and explain how they can sustain the platform's long-term value and adaptability to the Manga's evolving needs

Appendix 2. Customer Research

Business research on Manga

Competitive Analysis

The retail market is currently experiencing intense competition, driven by rapid shifts in consumer behavior, the rise of digital-native brands, and increasing cost pressures. To stay ahead, companies are leveraging data to enhance sales through personalization, optimize inventory levels through predictive analytics, reduce operational costs, and strengthen supply chain resilience. Manga has reported solid annual revenues of €280 million; however, its growth rate of 3.5% falls short of the market average. This gap is particularly evident among younger demographics and in digital channels, where Manga's engagement and sales growth remain underwhelming, with online sales increasing only 1.2% year-over-year.

Key Decision Makers

Marta Ríos (she/her) - Chief Operations Officer (COO): With 25 years at Manga and a place on the original founding team, Marta is a pragmatic leader who prioritizes results over ideology. She's open to disruptive change—but only if the business case is solid and the solution is end-to-end. Marta demands that any modernization proposal be comprehensive, controlled-risk, and demonstrably capable of improving efficiency, growth, and competitiveness before she gives her final approval.

Javier Medina (he/him) - Chief Technology Officer (CTO): Javier has been with Manga for 15 years and is the architect behind the current data infrastructure. He's skeptical of cloud migrations, viewing them as potentially costly and a threat to long-standing operational control. He champions open-source technologies and modular systems and is highly concerned about vendor lock-in. While not completely against change, he wants guarantees that modernization won't sacrifice flexibility or control.

Manuel Ortega (he/him) - Chief Financial Officer (CFO): Another founding team member with 25 years at Manga, Manuel has traditionally favored stable, long-term infrastructure partnerships and signed the current colocation deal. However, recent financial analysis has shown that Manga is spending twice as much as competitors while achieving only half the output. Manuel is increasingly concerned with operational inefficiency and is open to

modernization—if it can show a clear path to cost reduction, better ROI, and sustainable long-term value.

Laura Smith (she/her) - Information Security Manager: Laura is new to Manga and came in with a mandate to fix weak points in security and compliance. With a strong background in data regulation and lifecycle governance, she is laser-focused on ensuring that any future data architecture meets the highest standards of security, privacy, and legal compliance. After a major incident that led to the previous InfoSec Manager's dismissal, Laura views robust security as non-negotiable for any modernization effort.

Alex Lee (they/them) - Head of Data & AI: Alex recently joined Manga from a leading competitor, bringing strong experience with modern cloud analytics and DataOps. They see the current architecture as outdated and overly rigid, with failing ETL processes leaving the data outdated or incomplete, and limited access to siloed databases. Simply moving what they have to the cloud won't solve the problem. They need a modern, robust, and reliable architecture.

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Appendix 3. Sample Datasets

Here's a description of Manga's main datasets and their sources. Find sample files uploaded in the course platform:

Website Metrics: Captures user interactions on Manga's digital storefront. Each row logs a specific action—such as viewing a product, adding to cart, or completing a purchase—alongside timestamps, session IDs, optional customer IDs, product IDs, and order IDs (if a purchase occurred). This unstructured, event-based data originates continuously from web analytics platforms and digital tagging frameworks integrated into the e-commerce site and mobile applications.

Sample File: *web_metrics_sample.csv*

Customer Reviews: Collected from Manga's product feedback channels. It includes timestamps, customer and product identifiers, review text, star ratings, and associated image uploads. This data originates from the review system on the website or mobile app and is crucial for sentiment analysis, product improvement, and trust-building with new customers.

Sample File: *customer_reviews_sample.csv*

Inventory: Contains up-to-date stock levels of fashion products across different store locations. Each entry includes a product ID, store ID, current price, stock count, and a product description. It originates from Manga's inventory management systems (typically part of an ERP or POS platform) and supports supply chain visibility and pricing strategy.

Sample File: *inventory_sample.csv*

Sales: Records showing transactional data, including date/time, customer ID, product and store IDs, payment method, final purchase price, and applied discount. Prices and store-product relationships are consistent with the inventory system. This data comes from the point-of-sale (POS) system and e-commerce transaction logs and is essential for revenue tracking, promotions analysis, and customer behavior modeling.

Sample File: *sales_sample.csv*

Customers: Demographic information about Manga's customers, including full name, age, gender, address, postal code, and loyalty program status. This data originates from CRM systems, loyalty programs, and e-commerce registration, forming the foundation for segmentation, personalization, and customer lifecycle analysis.

Sample File: *customers_sample.csv*

External Factors: History of aggregated external contextual information that may impact sales: weather conditions, fuel prices, bank holidays, local events, marketing campaigns, and total daily sales and item volume. This data combines sources such as public APIs (weather, holidays), marketing calendars, and internal sales summaries, enabling demand forecasting and campaign evaluation.

Sample File: *external_factors_sample.csv*

Shipping: Product deliveries, containing order IDs, delivery addresses, estimated delivery time, delivery and actual shipping costs, and return status. The data is sourced from logistics providers and Manga's own fulfillment systems. It helps analyze delivery performance, cost efficiency, and return behavior.

Sample File: *shipping_sample.csv*